

Project Use Case Document

Project Title: Employee Lifecycle and Attrition Analytics Pipeline

Role: Data Engineer

Domain: Human Resources (HR) Analytics

1. Project Overview

The "Employee Lifecycle and Attrition Analytics Pipeline" is an end-to-end data engineering solution designed to process, track, and analyze employee records over time. By leveraging cloud data integration and a Medallion Architecture, this project transforms raw, fragmented HR data into clean, aggregated metrics to help organizations understand workforce dynamics and predict attrition risks.

2. Problem Statement

Modern HR teams face significant challenges in tracking the complete lifecycle of an employee. Data regarding promotions, department transfers, salary adjustments, and resignations is often fragmented across daily snapshots or isolated systems. Without a historical tracking mechanism, it is incredibly difficult to identify patterns in employee turnover, measure accurate retention rates, or flag departments with high attrition risks.

3. Proposed Solution

To solve this, we will build an automated data pipeline that processes daily batches of employee records. Instead of just overwriting old data, the system will track historical changes (such as a salary bump or a role change) while maintaining a current snapshot of the workforce. The final output will provide aggregated analytics on average tenure and monthly attrition rates, giving HR leadership actionable insights.

4. Complete Technical Architecture & Tech Stack

Below are all the technologies, tools, and concepts that will be utilized to build this project from scratch:

- **Programming Languages:**
 - Python: Used to generate the mock daily datasets (using Python basics and data structures).
 - SQL: Used for complex data transformations, filtering, and aggregating (utilizing JOINS, GROUP BY, CASE statements, CTEs, and Window Functions).
- **Cloud Storage & Ingestion (ETL Concepts):**
 - Azure Storage Accounts: To securely store the raw daily files.
 - Azure Data Factory (ADF): To build automated pipelines that move data from the source into the cloud storage.
- **Big Data Processing:**
 - PySpark: Used to read the raw files, handle DataFrames, and apply large-scale data transformations.
- **Workspace & Data Environment:**
 - Databricks: The main workspace and cluster environment for writing and running the code.
 - Delta Lake: The storage layer that allows us to update and manage the data reliably (ACID properties).
- **Data Modeling & Architecture:**
 - Medallion Architecture: Organizing data logically into Bronze (Raw), Silver (Cleaned/Tracked), and Gold (Business Ready/Aggregated) tables.
 - Slowly Changing Dimensions (SCD Type 2): Using Delta Lake MERGE operations to keep a historical log of whenever an employee's data changes.